Paper 3
NeuroAgentBench

A reproducible benchmark for evaluating AI agents on neuroimaging workflows, scientific validity, execution, and provenance

COMPLETE RESEARCH–TOOL–PUBLICATION BLUEPRINT | Version 1.0 | 25 October 2026

Planning document, not a completed experimental study. All benchmark sizes, success criteria, schedules and figures are proposed, not observed results.

Executive decision

Build a public benchmark suite and evaluator first; make scientific artifact validity—not "the agent said it succeeded"—the primary outcome. Preserve a frozen, independently adjudicated test set for the paper while a public development set supports adoption. There are two realistic publication tracks: a reusable dataset/data descriptor (Scientific Data or GigaScience) and a substantive research article about cross-agent reliability and failure modes (Nature Machine Intelligence or Nature Communications as conditional stretch targets). Avoid assuming one benchmark automatically supports two independent papers.

Decision	Recommendation
Working title	NeuroAgentBench: Evaluating AI Agents on Reproducible Neuroimaging Workflows and Scientific Validity
First release	Local CLI, Docker fixtures, 30–50 development tasks and report generator
Main paper asset	Versioned, expert-adjudicated benchmark with artifact-based scoring
Primary manuscript route	Benchmark/data-resource report; Scientific Data Data Descriptor if scope requirements are met
Stretch route	AI systems findings with multi-agent, cross-domain or broadly generalizable reliability evidence
Core caution	Do not claim domain-first or benchmark novelty until systematic search and gap mapping are completed

1. Research concept and significance

1.1 Problem definition

Scientific AI agents may write plausible analyses, choose tools, and execute programs, yet still produce anatomically misregistered images, mislabeled regions, invalid statistical interpretations, or non-reproducible outputs. Standard code benchmarks emphasize patch correctness; machine-learning benchmarks often emphasize predictive scores; many scientific-agent benchmarks span domains but do not deeply test neuroimaging-specific scientific contracts. NeuroAgentBench would evaluate end-to-end computational actions and resulting artifacts, with an explicit distinction between infrastructure success, technical validity, scientific validity, and warranted interpretation.

1.2 Key research questions

- RQ1: How accurately do generalist and specialist AI agents construct and execute valid neuroimaging workflows under identical tools and resource limits?
- RQ2: How frequently are technically successful runs scientifically invalid, and which error categories dominate?
- RQ3: How sensitive are outcomes to agent scaffold, model, documentation, tool access, feedback and execution budget?
- RQ4: Are benchmark judgments reproducible and concordant with blinded domain-expert assessment?
- RQ5: How well does performance transfer across imaging modalities, datasets, pipelines, and unseen error patterns?

1.3 Hypotheses to preregister

- H1: The technical-execution success rate exceeds the scientifically valid artifact success rate by a nontrivial margin.
- H2: Rich tool descriptions or access to deterministic scientific validators improve validity compared with plain shell/documentation conditions.
- H3: Agent rankings vary by scientific task category and execution environment.
- H4: Performance drops on unseen datasets, unusual metadata, and adversarially invalid workflow requests.
- H5: Explicit confidence/abstention and quality-control checks reduce scientifically unsafe false-success claims.

These hypotheses are testable proposals, not findings. Decide the primary hypothesis and minimally important effect before the main study.

1.4 Expected contributions

- A taxonomy of neuroimaging research-engineering tasks and failure modes.
- Containerized tasks with reference artifacts, valid acceptance criteria, and linked scientific rationales.
- A machine-readable evaluator distinguishing executable correctness from scientific correctness.
- Open baseline agent runs, budget logs, aggregate scores, and diagnostic failure analysis.
- An extensible protocol for external community tasks with explicit inclusion and expert-review standards.

2. Literature review: established work and remaining opportunities

This is a focused, source-linked preliminary literature map, not a systematic review. Review the complete papers, follow backward/forward citations and document the final search strategy before claiming originality.

Research stream	What already exists	Gap NeuroAgentBench should test

ScienceAgentBench (2025)	102 data-driven scientific tasks with expert validation; verified benchmark update in 2026 [1]	Neuroimaging-specific pipeline and artifact correctness; must compare scoring rigor rather than claim expert evaluation is new
SciAgentArena (2026)	~200 interactive, stepwise-verified tasks across five science domains [2]	Deep MRI/DWI/atlas/space/QC semantics; avoid claiming stepwise verification is new
MLR-Bench (2025)	201 open-ended ML research tasks, judge and research-stage evaluation [3]	Clinical imaging computation and objective artifact-level scientific validity rather than paper-quality scoring
MLE-bench (2024)	75 ML engineering competitions, resource controls and data contamination discussion [4]	Image processing workflow semantics and provenance
SWE-bench (2024)	Real software issues, Docker environments and fail-to-pass tests [5]	Scientific validity cannot be fully reduced to unit tests
MLRC-Bench (2025)	Objective research-competition evaluation [6]	Domain-expert scientific QC and task-specific medical imaging artifacts
Nipype / BIDS	Reusable neuroimaging workflows and data standards [7–9]	Agent behavior and error detection over established tools, not reinvention of workflows
Scientific data publication	Technical validation, public repository, reproducible data record [10]	Publish reusable, versioned task dataset and evaluator rather than marketing claims

2.1 What would NOT be novel

- A leaderboard for language models by itself.
- Docker execution and pass/fail checking by themselves.
- A set of biomedical prompts and expected answers without independent task validation.
- Use of human experts or stepwise verification by itself.
- A neural agent that merely calls Nipype, FreeSurfer, or BIDS Validator.

2.2 Candidate distinctive contribution

A rigorously adjudicated, artifact-level benchmark of neuroimaging agents in which a correct answer requires compatible imaging data semantics, anatomically appropriate spatial operations, robust QC, reproducible provenance, and conservative behavior when inputs are invalid. The novelty claim must be supported by a literature matrix and evidence that existing benchmark suites do not already cover the combination at comparable depth.

2.3 Search protocol for final review

Databases: PubMed, Google Scholar, Semantic Scholar, OpenAlex, arXiv, ACM Digital Library, IEEE Xplore and conference proceedings. Search blocks: (“AI agent” OR “LLM agent” OR “autonomous scientist”) AND (“benchmark” OR “evaluation”) AND (“neuroimaging” OR “MRI” OR “scientific workflow” OR “scientific validity”); also search benchmark names individually. Include technical reports/preprints but identify their peer-review status and versions. Screen 2023–October 2026 plus foundational neuroimaging workflows. Track title, DOI,

publication status, tasks, agent access, ground truth, scoring, costs, error categories, limits and overlap. Rerun immediately before submission.

3. The public tool: NeuroAgentBench

3.1 Users and value

User	Primary value
Agent researchers	Evaluate scientifically meaningful behavior rather than claims alone
Neuroimaging labs	Assess whether agents are reliable for familiar pipeline tasks
Tool/agent developers	Regression tests for pipeline integrations and validation layers
Students and educators	Sandbox exercises in scientific workflow QA
Platform maintainers	Inspect systematic agent failure modes without private patient data

3.2 MVP

- neuroagentbench list: enumerate task IDs, versions, tags and resource requirements.
- neuroagentbench run --agent adapter.yaml --suite dev: run against isolated fixtures.
- neuroagentbench evaluate --runs runs/: apply objective validators and emit structured reports.
- neuroagentbench report --format html/json: visualize pass rates, failure categories and costs.
- Container recipe for each task with deterministic seed and reference artifact hashes.
- Developer documentation, test fixture authoring guide, benchmark changelog and versioned examples.

Commands above are prospective design examples; they have not been implemented or tested.

3.3 Architecture

Task registry → sandbox/fixture manager → agent adapter (shell/API/MCP) → constrained execution environment → artifact collector → scientific and technical validators → results database → HTML dashboard and publication analysis pipeline. Preserve full audit records with secrets scrubbed and no upload of sensitive patient data.

3.4 Core task record schema

A task specification must include: task_id, benchmark_version, domain/subcategory, natural-language request, allowed tools, starting data manifest and licenses, prerequisites, time/memory budget, expected artifacts, execution oracle, scientific oracle, prohibited actions, uncertainty/abstain criteria, severity tags, scoring version, expert provenance, and review status. Keep hidden evaluator criteria separate from publicly exposed prompts when appropriate.

3.5 APIs and integration

Use Python 3.11+, pytest, Docker or Apptainer; typed Pydantic task models; structured JSONL run events; pluggable adapters for shell, MCP, and function-calling agents. Prefer CLI-first offline reproducibility and add hosted submissions after securing isolated compute, abuse prevention, artifact scanning, and budget controls. Provide an SDK allowing researchers to contribute tasks and validators.

3.6 Release governance

- Open-source software license and explicit task/dataset reuse license; verify compatibility with upstream licenses.
- Contributor guide, code of conduct, responsible disclosure and security policy.
- Versioned public development set; frozen evaluation set with leakage controls.
- Public model results should identify exact model snapshot, scaffold, prompts, tools, seeds and budget.
- Community task submissions require duplicate screening, expert review and tests before inclusion.
- Separate public alpha usability metrics from prespecified paper outcomes.

4. Task taxonomy and benchmark construction

Category	Example tasks	Objective checks
BIDS and acquisition metadata	Repair malformed sidecar or dataset layout	BIDS Validator + expert interpretation of metadata
Anatomical segmentation	Configure T1 pipeline and inspect segmentation output	File structure, regional plausibility, segmentation QC
Image registration	Align anatomical labels and DWI reference	Transform direction, image grid, overlap, orientation
Diffusion preprocessing	Choose compatible preprocessing for available acquisitions	Inputs, gradients, b-values, preprocessing artifacts
Tractography/connectomics	Construct atlas-specific matrices	Atlas correspondence, symmetry, dimensions, edge semantics
HPC diagnosis	Analyze Slurm errors, memory and container failures	Root-cause fixture and acceptable remediation
Statistical/provenance	Reproduce metric table and processing provenance	Numerical tolerances, artifact provenance
Safety/abstention	Reject invalid or impossible scientific request	Correct abstention or justified escalation

4.1 Proposed task counts

Stage	Scope	Purpose
Pilot	30–50 tasks, 5–8 categories	Check task feasibility, ambiguity, runtimes and scoring agreement
Main v1	150–250 tasks (conditional)	Broad enough for stable subgroup estimates if justified by pilot
External holdout	~20–30% independently sourced when feasible	Test transfer to unseen datasets, errors and workflows
Hidden evaluation	Subset reserved from developer access	Reduce tuning and leaderboard gaming

These are planning ranges. Final task counts require precision analysis for primary outcomes, annotation capacity, compute budget, and coverage goals.

4.2 Dataset strategy

Tier 1: synthetic BIDS manifests, toy images, labels, tractograms, transform matrices and controlled corruptions; ideal for exact scoring. Tier 2: selected OpenNeuro datasets with fixed release snapshots, DOI/accession, participant/session identifiers and license checks. Tier 3: any approved institutional datasets only as restricted

external validation with appropriate governance; never needed for the open benchmark. BIDS examples include structural layouts but may contain placeholder/empty image files; do not run real segmentation on those fixtures. [8,11]

4.3 Annotation protocol

- Draft tasks from independently described real workflows and plausible failure modes.
- Two blinded subject-matter experts review validity, expected outputs, severity and alternatives.
- Disagreements are adjudicated by a third expert with decisions recorded.
- Annotate possible valid solution sets, not a single arbitrary script.
- Require executable reference solutions or justified human-graded rubric.
- Pilot inter-rater reliability (e.g., agreement and Cohen kappa or Krippendorff alpha as appropriate).
- Reject ambiguous tasks until clarified; record excluded tasks and reasons.

4.4 Example task specifications

- T01: Given T1w anatomical labels and a DWI-space tractogram, construct the valid intermediate steps; fail direct-space misuse.
- T02: Given phase-encoding metadata that are insufficient to infer an unambiguous preprocessing correction, identify uncertainty rather than invent metadata.
- T03: Given a failed Slurm log with out-of-memory symptoms, propose a resource-aware rerun preserving scientific settings.
- T04: Given valid matrix dimensions but mislabeled DK/DKT region orders, detect atlas semantic mismatch.
- T05: Given a segmentation report with incomplete stage outputs, refrain from treating missing regions as biological zero volume.

5. Scoring and scientific validity

Use a multi-dimensional score rather than one opaque composite. Primary endpoint candidate: fraction of tasks producing a scientifically valid, reproducible final artifact under a fixed budget. Secondary endpoints: technical execution, scientific constraint pass, provenance completeness, valid abstention, harm-weighted false acceptance, computational efficiency and repeatability.

Measure	Operational definition	Notes
Execution success	Required command/artifact produced and pipeline finishes	Not sufficient for scientific validity
Scientific validity	All mandatory science-specific checks pass	Ground truth independently reviewed
False success	Agent declares success but scientific oracle fails	Central scientific risk indicator
Abstention calibration	Correct refusal/escalation on impossible or invalid cases	Avoid rewarding blanket refusal
Reproducibility	Independent rerun gives accepted artifacts within tolerance	Account for nondeterministic algorithms
Budget efficiency	Time, CPU/GPU, tokens, monetary API spend	Normalize for environment
Provenance completeness	Required versions, parameters, inputs, seeds and transforms recorded	Check machine-readable fields
Review concordance	Automatic score agrees with	Audit exceptions

	blinded experts

5.1 Oracle design

Use layered grading: schema/execution checks; objective numerical/image criteria (where scientifically defensible); domain-specific semantic rules; blinded human adjudication for unresolved cases. Output accepted/warning/rejected/not-assessable per criterion. Do not use an LLM judge as sole arbiter of scientific correctness. Rubrics must describe acceptable numerical tolerances and alternative valid workflows.

5.2 Severity-aware evaluation

Group failures into low-consequence formatting problems, operational failures, major scientific incompatibilities, and critical invalid conclusions. Report per-class false-negative rates. Any severity-weighted aggregate must have a transparent predetermined weighting scheme; report unweighted outcomes too.

5.3 Fairness and contamination

Record model training cutoff where known, pre-existing exposure to benchmark tasks, retrieved-web permission, agent tool parity, scaffolds and prompts. Avoid adapting hidden tasks based on leaderboard performance; rotate or add future holdouts with published change logs. Distinguish model comparisons from system comparisons (different scaffolds).

6. Experiments and baselines

6.1 Controlled comparison matrix

Experiment	Conditions	Main outcome
E1: Baseline capability	Shell + documentation; structured tools; reference non-agent execution	Scientific artifact validity
E2: Agent diversity	At least 3 model families × 1–2 fixed scaffolds if resources allow	Between-agent heterogeneity
E3: Tool enrichment	Ordinary tool metadata vs SciCap-style semantics	Change in invalid configuration
E4: Preflight checking	Without vs with SciGuard feedback	Change in false success
E5: Difficulty shift	Known templates vs held-out tasks/datasets	Generalization gap
E6: Repeated trials	3–5 runs for selected task-agent conditions	Variance and stability
E7: Evaluation quality	Automatic oracle vs independent blinded expert rating	Validity of benchmark scoring

6.2 Experimental controls

- Fix dataset version, container digest, software versions and expected numerical tolerance.
- Use identical task instructions and access permissions for matched comparisons.
- Prespecify time, token, dollar, CPU, memory and retry ceilings.
- Randomize task order and record seeds; avoid cross-task filesystem contamination.
- Preserve raw events, intermediate outputs, agent plans and validator results for audit.

- Account for cached preprocessed images where full processing would be prohibitively expensive; document which tasks are real execution vs simulated fixture tests.

6.3 Statistical analysis plan

Primary unit: task, with repeated runs nested within task and agent. Present point estimates and 95% confidence intervals for valid-artifact rates, with task-stratified or hierarchical bootstrap when appropriate. For matched binary conditions use paired tests (e.g., McNemar) or mixed-effects logistic models with task-level effects. Investigate task difficulty and failure categories as planned moderators. Correct or explicitly label exploratory multiple comparisons. Report absolute risk differences and confidence intervals rather than only p-values. Predefine missing/timeout handling and safety-metric denominators.

6.4 Pilot and sample-size rationale

A 30–50 task pilot can estimate test instability, annotation burden and preliminary event rates; it is not powered by default. For the main study, calculate sample size from the minimally important difference in scientific-validity success between paired conditions and expected discordance; consider multiple repeated samples and group-clustered tasks. Publish the calculation script with assumptions.

6.5 Failure analysis

Inspect a stratified sample of success, false-success, false-rejection, and timeout cases. Code root causes: misunderstanding scientific input, invalid tool choice, coordinate/atlas issue, missing metadata, environment failure, statistical interpretation error, inadequate QC, hallucinated output, premature success, and appropriate abstention. Dual-annotate a subset and report disagreements.

7. Reproducibility, safety and ethics

- Use public or synthetic non-identifiable inputs for open releases; do not embed patient data, private HPC paths, access tokens or institution-restricted logs.
- Network off by default inside task containers; constrained file mounts; process isolation, timeouts, memory/CPU quotas, no privileged execution.
- Maintain signed/checksummed artifact manifests, seeds, versioned tool containers and benchmark release hashes.
- Publish scripts to rebuild the environment, verify reference outputs and regenerate paper tables/figures.
- Do not claim outputs are clinical diagnostic devices or use patient-care workflows without appropriate approvals.
- Assess institutional intellectual-property obligations before incorporating URMC-origin code or assets.
- Require consent and any appropriate ethical review for planned human-subject usability studies, separate from automated benchmark data.

8. Manuscript planning: two different publication paths

8.1 Path A: benchmark resource / Scientific Data Data Descriptor

Proposed title: “NeuroAgentBench: An expert-validated dataset of reproducible neuroimaging workflow tasks for evaluating AI agents.” The paper should describe dataset creation, annotation, file structure, access, technical validation and reuse. It must not be disguised as a hypothesis-testing research results paper; detailed model comparison and performance interpretation belong in a separate research article when scientifically distinct. Current journal guidance requires anonymously accessible data at initial review and formal public repository availability subsequently. [10]

1. Abstract (~170 words maximum per current journal recommendations, verify at submission).
2. Background & Summary: evaluation problem, dataset reuse need, comparison to available resources.
3. Methods: task sources, datasets, transformations, expert annotation, benchmark versioning, ethics.
4. Data Records: schema, field definitions, file structure, DOIs and repository contents.
5. Technical Validation: validity, grading consistency, reference execution, expert agreement and quality controls.
6. Usage Notes: installation, example commands, limits, contamination, hardware and intended scope.
7. Data/Code Availability, contributors, competing interests, funding, references.

8.2 Path B: AI systems / methods research article

Proposed title: “Scientific correctness is not execution success: A systematic evaluation of AI agents on neuroimaging workflows.” Target only if experiments reveal substantive findings and strong methodological generality. Introduction → Related Work → Benchmark & Evaluation Framework → Experimental Design → Results → Failure Taxonomy → Ablation/Generalization → Discussion → Limitations → Data/Code Availability.

8.3 Working abstract scaffold (no fabricated findings)

Background: Scientific agents increasingly operate complex analysis tools, but general benchmarks may not distinguish successful execution from scientifically valid outputs. Methods: We construct NeuroAgentBench, a task suite covering BIDS validation, anatomical processing, registration, diffusion and connectomics, HPC troubleshooting and provenance, with expert-reviewed reference solutions and artifact-level checks. Evaluation: We compare agents under controlled tool and resource budgets, measure validity, false-success, abstention, reproducibility and cost, and assess scorer agreement with blinded expert review. Results: [INSERT MEASURED COUNTS, ESTIMATES, CI, KEY FAILURE TYPES]. Conclusion: [STATE ONLY CONCLUSIONS WARRANTED BY RESULTS].

8.4 Required figures and tables

Figure or table	Contents
Figure 1	Benchmark design, task environments and scorer pipeline
Figure 2	Task taxonomy, public/hidden split and difficulty
Figure 3	Validity versus technical execution across agents
Figure 4	Failure modes, severe false successes and abstention
Figure 5	Ablation/generalization and cost–quality tradeoffs
Table 1	Comparison against ScienceAgentBench, SciAgentArena, MLR-Bench, SWE-bench
Table 2	Dataset/task inventory and rights
Table 3	Results, confidence intervals and paired comparisons
Supplement	Task review form, oracle rules, prompts, artifacts, exact environments

9. Journal tiers: conditions for credible targeting

Journal targets below are editorial-fit and research-ambition judgments—not impact-factor rankings or acceptance predictions. Recheck submission policies and article types immediately before submission.

Tier / journal	Best fitting manuscript	What must be demonstrated

A+ stretch: Nature Machine Intelligence	Broad scientific-agent reliability / evaluation research	Generalizable AI insights; technically sound evaluation; multiple agent systems; strong novelty versus ScienceAgentBench/SciAgentArena; external validation and conceptual significance beyond neuroimaging
A stretch: Nature Communications	Major biomedical or AI-for-science methods advance	Important advance for specialists with rigorous reproducibility; robust validity evaluation and compelling cross-setting finding
A strong data target: Scientific Data	Expert-validated task/data resource	Public data deposited and technically validated; useful reusable dataset; complete Data Records, metadata, methods and Usage Notes; avoid substantive hypothesis-results narrative
A/B strong specialist: GigaScience	Open scientific benchmark and workflow infrastructure	Transparent open data/code, workflow verification, repeatability and clear reuse contribution
B specialist: Imaging Neuroscience	Imaging-focused benchmarking/methodology	Strong neuroimaging methodological scope, real workflows, evaluated validity, openly reusable artifacts
Parallel conference track: NeurIPS Datasets & Benchmarks	Benchmarks with novel evaluation methodology	Clear distinct benchmark contribution, rigorous baselines, documentation, responsible data, novel insights and deadline adherence

9.1 Journal-specific gate: Nature Machine Intelligence

- Multi-agent study with stable high-quality implementations, ideally broader than one neuroimaging lab.
- A substantive new conclusion about intelligent-agent reliability or a reusable methodological evaluation advance.
- Strong prior-art positioning against SciAgentArena, ScienceAgentBench, MLR-Bench and other 2026 work.
- Blinded scientific evaluation, controlled experiments, ablations, held-out tasks, uncertainty estimates and transparent failure cases.
- Evidence that benchmark conclusions generalize across at least several tasks, tools, architectures and ideally additional scientific settings.

9.2 Journal-specific gate: Scientific Data

- A clearly reusable data product with public/downloadable task corpus and scoring metadata.
- Persistent repository and formal versioned DOI; license analysis for every source.
- Comprehensive data records, validation checks, acquisition/curation steps and expert annotation evidence.
- Technical validation of dataset quality, not a research manuscript arguing model superiority.
- Follow current reviewer-access, public deposition, code and ethical policies.

9.3 Decision rule

Choose the Data Descriptor if the strongest output is the open benchmark and its technical quality; choose a methods/AI journal if the independent scientific finding is a broadly meaningful advance. A separate later paper is justified only with distinct research questions and largely nonredundant analyses. Consult potential editors with a brief abstract and evidence summary once preliminary results exist.

10. Project implementation roadmap

Phase	Deliverables	Go / no-go gate
Weeks 1–3	Search protocol, novelty matrix, collaborators, repository structure	No unsupported “first benchmark” claims
Weeks 4–6	Task schema, 10 deterministic fixtures, evaluator prototype	Reference outputs repeatably scored
Weeks 7–10	30–50 task pilot; blinded review; 2 baseline agents	Scoring ambiguity and expert disagreements resolved
Months 3–5	Full task corpus v1, external task contributors, freeze split	Adequate coverage and validation evidence
Months 5–7	Multi-model evaluation; repeated runs; analysis pipeline	Predefined primary endpoint and budget achieved
Months 7–9	Independent audit, public package and primary manuscript	Release artifacts regenerate tables and figures
Months 9–12+	Journal submission and follow-on externally validated study	Paper novelty and publication route justified

Timing is illustrative and depends on engineering effort, compute, collaborators, governance and the size of the validated task set.

11. Personnel, infrastructure and estimated resource categories

Need	Minimum viable arrangement	Status to verify
Technical lead	Research engineer coordinating tasks, infra and reports	Available experience; protected effort needed
Domain experts	At least 2 scientific reviewers plus adjudicator	Recruit collaborators
Methods/statistics review	Design and CI/mixed-effects expertise	Recruit or consult
Engineering environment	Linux/Docker/Apptainer, Python tests and CI	Access and permissions to confirm
Compute	CPU analysis; selective GPU/preprocessing; bounded API use	Budget a pilot before main runs
Data	Synthetic and OpenNeuro release snapshots	Select accession/version/licenses
Repository/archive	GitHub + DOI-capable archive such as Zenodo	Create and maintain releases
Publication funding	Optional APCs, conference travel, annotation	Verify grants/waivers/institutional support

12. Anticipated peer-review criticisms

Likely critique	Required response
“Just another science-agent benchmark”	Show comparative matrix and new validity-grounded tasks absent from prior work
“Only one lab and one modality”	Multi-tool/multi-dataset design, external reviewers, held-out evaluations
“Your oracle rejects valid alternatives”	Multiple acceptable solutions, independent expert arbitration and audited false negatives
“Agents are not comparable”	Fixed scaffolds, controlled budgets, exact versions, distinction of model and system comparisons
“Data contamination undermines ranking”	Frozen holdouts, provenance and contamination audits, conservative interpretation
“No clinically relevant consequence shown”	Use controlled examples where invalid spatial/methodological decisions alter outputs; no clinical claims
“Not reproducible due to model drift”	Exact model IDs/dates, archived predictions, seeds and versioned environments
“User adoption proves validity”	Separate telemetry from controlled experiments and paper outcomes

13. Submission-ready checklist

- Literature search logged with date, inclusion/exclusion rules, screened prior work and novelty matrix.
- Task taxonomy documented; tasks validated independently by at least two experts.
- Public source dataset identifiers, licenses, artifact hashes, and exclusion of sensitive data documented.
- Executable oracles checked against reference solutions and alternative acceptable workflows.
- Main research protocol, primary endpoint, sample-size assumptions and study analysis committed before final runs.
- Baseline configurations, model snapshots and budgets frozen.
- All experiment results archived; confidence intervals, missing cases and ablations reported.
- Public dataset and code deposited per target journal requirements with DOI and documentation.
- Manual spot audit verifies figures/tables from archived machine-readable results.
- Manuscript distinguishes benchmark construction from methodological hypothesis testing and discloses limitations.
- Authorship, attribution, IP and institutional permissions confirmed.
- Target journal instructions, APCs, ethics, and AI-use declarations rechecked before submission.

14. First six concrete actions

8. Create repository neuroagentbench with LICENSE, README, CITATION.cff, task-schema JSON and contributor instructions.
9. Build five small fully deterministic tasks: BIDS filename error, coordinate-space mismatch, missing transform, atlas label inconsistency, and Slurm resource diagnosis.
10. Implement artifact/report validators for those five tasks; produce golden reference outputs.
11. Recruit two independent neuroimaging annotators; pilot annotation form and disagreement resolution.
12. Run a fixed agent baseline and a deterministic human/reference pipeline on the same tasks with logged budgets.

13. Draft a 2-page prespecified protocol and complete comparison matrix against ScienceAgentBench and SciAgentArena before adding new tasks.

15. Selected literature and authoritative resources

[1] Chen et al. ScienceAgentBench: Toward Rigorous Assessment of Language Agents for Data-Driven Scientific Discovery. ICLR 2025.. https://hal.cs.princeton.edu/scienceagentbench.

Relevance: Closest expert-curated scientific-agent benchmark; 2026 evaluation verification update matters.

[2] Liu et al. Benchmarking AI Agents for Addressing Scientific Challenges Across Scales (SciAgentArena). 2026 preprint.. https://arxiv.org/abs/2606.12736.

Relevance: Closest interactive multi-domain scientific benchmark; stepwise scoring and weak validity checking.

[3] Chen et al. MLR-Bench: Evaluating AI Agents on Open-Ended Machine Learning Research. NeurIPS Datasets and Benchmarks 2025.. https://proceedings.neurips.cc/paper_files/paper/2025/hash/ab8dd000d6f87f40061a73f8bca7fae4-Abstract-Datasets_and_Benchmarks_Track.html.

Relevance: Research-stage evaluation and issues with fabricated experimental results.

[4] Chan et al. MLE-bench: Evaluating Machine Learning Agents on Machine Learning Engineering. 2024 preprint.. https://arxiv.org/abs/2410.07095.

Relevance: Resource-controlled agent engineering evaluation and contamination concerns.

[5] Jimenez et al. SWE-bench: Can Language Models Resolve Real-world GitHub Issues? ICLR 2024.. https://proceedings.iclr.cc/paper_files/paper/2024/hash/edac78c3e300629acfe6cbe9ca88fb84-Abstract-Conference.html.

Relevance: Executable real-world task fixtures and verified grader design.

[6] Zhang et al. MLRC-Bench: Can Language Agents Solve Machine Learning Research Challenges? 2025 preprint.. https://arxiv.org/abs/2504.09702.

Relevance: Objective evaluation of research methods rather than subjective agent narration.

[7] Gorgolewski et al. Nipype: A Flexible, Lightweight and Extensible Neuroimaging Data Processing Framework in Python. 2011.. https://pmc.ncbi.nlm.nih.gov/articles/PMC3159964/.

Relevance: Established workflow representation, execution and interfaces; avoid duplicated novelty.

[8] Gorgolewski et al. The Brain Imaging Data Structure (BIDS). Scientific Data 2016.. https://www.nature.com/articles/sdata201644.

Relevance: Canonical neuroimaging data organization and metadata.

[9] Gorgolewski et al. BIDS Apps: Improving Ease of Use, Accessibility, and Reproducibility of Neuroimaging Data Analysis Methods. PLOS Computational Biology 2017.. https://pubmed.ncbi.nlm.nih.gov/28278228/.

Relevance: Established portable neuroimaging application pattern.

[10] Scientific Data. Submission guidelines and Data Descriptor template.. https://www.nature.com/sdata/submission-guidelines.

Relevance: Current data deposition, Data Records, Technical Validation and manuscript requirements.

[11] OpenNeuro public BIDS data repository.. https://openneuro.org/.

Relevance: Public reproducible real-data fixture acquisition; record specific accession and dataset snapshot.

[12] BIDS Examples project.. https://github.com/bids-standard/bids-examples.

Relevance: Metadata and structure fixtures; not necessarily real image content.

[13] Nature Machine Intelligence aims and scope.. https://www.nature.com/natmachintell/aims.

Relevance: Conditional elite AI-research journal fit.

[14] Nature Communications aims and scope.. https://www.nature.com/ncomms/ncomms/aims.

Relevance: Significance-to-specialists threshold for major broad research findings.

[15] GigaScience Data Note instructions.. https://academic.oup.com/gigascience/pages/data_note.

Relevance: Secondary benchmark/data-note route.

[16] SciAgentArena public project.. https://sciagentarena.github.io/.

Relevance: Open evaluation resources and baseline implementation ideas.

[17] OpenAI MLE-bench code repository.. https://github.com/openai/mle-bench.

Relevance: Examples of evaluator and model submission handling; current leaderboard handling changes over time.

[18] ScienceAgentBench verified-release documentation.. https://github.com/OSU-NLP-Group/ScienceAgentBench.

Relevance: Versioning lesson: prevent false-negative grading and re-evaluate affected submissions.

Appendix A. Example machine-readable task (illustrative)

task_id: nab-register-001

category: spatial-registration

input: T1-space atlas labels + DWI-space tractogram metadata

request: produce valid pipeline plan to construct an atlas-labeled connectome

required_behavior: identify spatial mismatch, request/validate transformation, prohibit direct use

expected_outputs: structured plan + transformation provenance + QC report

oracle: reject direct-space misuse; accept correctly justified compatible alternatives

safety: public/synthetic inputs only; no unauthorized external network

resources: CPU-only planning; max runtime and retry limits in task metadata

Appendix B. Example result table shell (no results claimed)

Agent / condition	N tasks	Technical pass %	Scientific valid %	False success %	95% CI
Baseline A	TBD	TBD	TBD	TBD	TBD
Baseline B	TBD	TBD	TBD	TBD	TBD
Agent + semantic descriptions	TBD	TBD	TBD	TBD	TBD
Agent + semantic validator	TBD	TBD	TBD	TBD	TBD

Final note: Literature is a focused starting bibliography, not a claim that all relevant competing methods have been captured. Journal tiers are aspirational. No agent experiments have been conducted as part of this document.